

Observer models — step-by-step comparison

A single-view comparison of the observer models built for this project, aligned stage by stage. Each model's own equation labels are given in full in its individual reference:

Model	File	Equation labels
Switch — paper's Switching observer (Laquittaine & Gardner 2018)	switching_observer.py	H1–H5
HB - Rachel — our hb_integration observer (the abstract's model)	hb_integration.py	M1–M5
HB - Salma — independent hierarchical observer	hierarchical_confidence/model.py	B1–B5
HB - Rommana — reliability-mixture observer	reliability_mixture_model.py	R1–R5

The models do not number 1:1 (the Switch keeps evidence and prior as two separate read-outs; the hierarchical models integrate them; the reliability mixture blends two fixed-location components), so the tables below align on **conceptual stages** rather than on each model's internal step number.

The three "HB" models all sit on the SAME hypothesis family — learn something about the prior trial-by-trial — but they disagree on **what** is learned: HB - Rachel and HB - Salma learn the prior *concentration* κ (holding the mixing behaviour fixed/derived); HB - Rommana learns the prior *reliance* (the mixture weight itself) and holds κ fixed per block.

Stage-by-stage comparison

Stage 1 — Prior representation

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Form	Delta at 225° (prior read-out)	$\alpha \cdot V(\theta; 225, \kappa) + (1-\alpha)/360$	Pure $V(\theta; 225, \kappa)$, per κ	Pure $V(\theta; 225, k_prior)$ component
Mixture floor (uniform)	via lapse only	yes, weight $(1-\alpha)$	no ($\kappa=0$ grid point only)	no (floor via lapse only)
Prior width set by	fitted k_prior per block	learned belief $b_t(\kappa)$	learned belief $H_t(\kappa)$	fitted k_prior per block
Label	H2	M1	B1	R1

Stage 2 — Sensory / likelihood component

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Form	$V(\cdot; \theta_true, k_like(c))$	$V(\cdot; \theta_true, k_like(c))$	$V(\cdot; \theta_true, k_sensory(c))$	$V(\cdot; \theta_true, k_llh(c))$
Measurement noise	marginalised (Girshick)	marginalised (Girshick)	marginalised (Girshick)	none — component centred on true dir
Per-coherence reliability	3 k_like	3 k_like	3 $k_sensory$	3 k_llh
Label	H1	M2	B2 (grid)	R2

Stage 3 — How prior and evidence are combined (the defining axis)

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Rule	MIXTURE — commit to prior OR evidence per trial	INTEGRATE, then average read-outs	INTEGRATE into one prior, then read out	MIXTURE — two fixed-location components
Formula	$P_e \cdot evidence + P_prior \cdot prior + P_rand \cdot unif$	$\Sigma_k b(\kappa) \cdot R_k$ (avg of per- κ read-outs)	one posterior from $\Sigma_k H(\kappa) \cdot V$, then MAP	$w \cdot V(225, k_prior) + (1-w) \cdot V(\theta_true, k_llh)$
κ marginalised...	n/a (no belief)	AFTER read-out	BEFORE read-out	n/a (no belief over κ)
Bimodality is...	imposed by stochastic commit	emergent from uniform floor	emergent from integrated posterior	imposed by the two-location mixture

Label	H3+H4	M4	B3+B4	R3
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Stage 4 — Reliance-on-prior weight

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
What sets it	derived ratio $k_{\text{prior}}/(k_{\text{prior}}+k_{\text{e}})$ (Eq. 6)	fitted α (fixed across blocks)	emergent from belief (no explicit weight)	learned online — $\text{prior_reliance } w_t$
Free parameter?	no (derived)	yes (α)	no	the <i>learning rate</i> α is fitted; w_t itself is dynamic
Label	H3	M1	—	R3/R4

Stage 5 — Learned latent (trial-by-trial dynamics) ← where the HB models diverge

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Learns across trials?	no — k_{prior} fixed per block	yes — belief $b_t(\kappa)$	yes — belief $H_t(\kappa)$	yes — mixture weight w_t
WHAT is learned	—	prior concentration κ	prior concentration κ	prior reliance (mixture weight)
Update rule	none	forget then Bayes-correct on feedback	forget then Bayes-correct on feedback	delta rule $w \leftarrow w + \alpha \cdot (\text{agreement} - w)$
Driven by	—	feedback direction	feedback direction	5-trial smoothed feedback vs prior mean
Reset	n/a	no (carried)	no (carried)	at session boundaries only
Label	—	M5	B2	R4

Stage 6 — Read-out type

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Read-out	MAP (Girshick push-forward)	MAP per κ (Girshick)	MAP, tie-aware	none — percept IS the mixture density
Tie handling	argmax	argmax	ties share mass 1/#ties	n/a (no MAP step)
Label	H1/H2	M3	B4	R3

Stage 7 — Motor noise + lapse

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Motor	■ $V(\cdot;0,k_{\text{motor}})$	■ $V(\cdot;0,k_{\text{motor}})$	■ $V(\cdot;0,k_{\text{motor}})$ via FFT	■ $V(\cdot;0,k_{\text{motor}})$ via FFT
Lapse	renormalised into Eq. 6 weights	(percept + $p_{\text{random}} \cdot \text{unif}$)/(1+ p_{random})	(1–lapse)·percept + lapse/ n_{θ}	(1–lapse)·percept + lapse/360
Label	H5	M4	B5	R5

Numerical grid & fitting

Aspect	Switch	HB - Rachel	HB - Salma	HB - Rommana
Direction grid	360 (1°)	360 (1°)	72 (5°)	360 (1°)
κ grid	n/a	~15 points	16 points (incl. $\kappa=0$)	n/a (no belief over κ)
Compute	linear space	linear space	log space (logsumexp)	linear space
Optimiser	(fitter-dependent)	(fitter-dependent)	Powell + Latin-hypercube multistart	per-subject NLL minimisation

Parameter counts

Model	N	Parameters
Switch	9	3 k_like + 4 k_prior + k_motor + p_random
HB - Rachel	7	3 k_like + α + k_motor + p_random + λ
HB - Salma	6	3 k_sensory + p + k_motor + lapse
HB - Rommana	10	3 k_llh + 4 k_prior + α + k_motor + lapse_rate

Note that α means different things across models. In HB - Rachel it is the fixed prior *mixture weight*. In HB - Rommana it is the *learning rate* of the delta rule that updates the mixture weight — the mixture weight itself (prior_reliance) is a dynamic latent, not a fitted parameter. HB - Rommana keeps the Switch's four fixed per-block k_prior and adds the learned reliance on top, which is why it is the most heavily parameterised (10).

One-line summary of the differences

- **Switch vs the three HB models:** the Switch's prior width is a *fixed* fitted constant per block with no trial-by-trial dynamics. All three HB models learn *something* about the prior online.
- **What each HB model learns (the real fork):**
- **HB - Rachel** and **HB - Salma** learn the prior *concentration* κ (how tight the bump is), holding the mixing behaviour fixed or derived. They differ from each other in numerics and combination rule (Rachel averages per- κ read-outs *after*; Salma integrates into *one* prior *before*; Rachel keeps a fitted α and forgets linearly, Salma has no α and forgets geometrically). Empirically they agree near the prior and diverge in the far-from-prior bimodality regime (mean total-variation ≈ 0.16 far, ≈ 0.06 near).
- **HB - Rommana** learns the prior *reliance* (the mixture weight itself) via a delta rule against smoothed feedback, and holds κ fixed per block. Its bimodality is *imposed* by a genuine two-location mixture (like the Switch), not *emergent* from a Bayesian posterior (like Rachel/Salma) — so it is, in effect, "the Switch with an online-learned, continuous switch weight."
- **A known caveat for HB - Rommana** (from its own README): k_prior becomes unidentifiable whenever prior_reliance collapses toward zero in a condition (chiefly the 80° block), because the same k_prior shapes both the percept mixture and the feedback-agreement term — a dual role worth resolving before relying on fitted k_prior values.